

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Chloroplasts (1) unevenly thickened cell walls (1)	2			2		
	(b)	(i)		Expressed to 3sf = 198 = 4 marks If incorrect sig fig allow 3 marks for 198.159 / 198.0758/ 198.16 If rounded area allow 3 marks for 197 / 200 If answer incorrect award 2 mark for calculation of area 0.07065 / 0.07068/ 0.071 If area incorrect award 1 mark for Substitution into formula ($\pi \times 0.15 \times 0.15$) = 1 mark *if candidate uses diameter instead of radius award 3 marks for 49.5 Award 2 marks for incorrect sig fig 49.539/ 49.522 Award 1 mark for calculation of area 0.2826 / 0.2827		4		4	4	4
		(ii)		Epidermal cells are larger {in shade/ diagram 2}/ORA (1) so fewer of them (per unit area) {in shade/ diagram 2}/ ORA (1)			2	2		
	(c)	(i)		Prevent {air/ bubbles} entering xylem (1) which would break {cohesion/ hydrogen bonds}/ affect cohesive forces (between water molecules) (1)	2			2		2

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		A. Humidity falls (1) B. because {ABA/ hormone} causes closure of stomata (1) C. causes reduced {transpiration/ water vapour loss/ evaporation} (from leaf/ stomata)(1) NOT stops D. (Benefit to plant) conserves water / reduces water loss / maintain turgor/ prevent wilting (when low soil water) (1) NOT stops Penalise stop once only		2	2	4		
				Question 7 total	4	6	4	14	4	4